

COMPUTER VISION PROJECT

Feature-Based Image Matching and Automatic Panorama Construction

Feature detection, description, matching, RANSAC, homography estimation and mosaicking

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Repository: github.com/Qwamzz/Computer-Vision-Projects

Implementation: Python 3.13.14 · OpenCV 5.0.0 · NumPy 2.5.2

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Summary

A complete classical panorama pipeline (feature detection, description, matching, RANSAC outlier rejection, homography estimation, warping and blending) was implemented in Python and OpenCV and evaluated quantitatively against exact ground truth. The descriptor matcher, the normalised DLT homography estimator and the RANSAC loop are implemented from first principles; OpenCV is used for the detectors under study and, deliberately, as an independent check on those implementations (the hand-written matcher agrees with `cv2.BFMatcher` on 100 % of correspondences, and the hand-written homography agrees with `cv2.findHomography` to 0.002 px).

Three detector/descriptor configurations were compared. SIFT gave the most accurate alignment (0.070 px mean error against ground truth); ORB detected and described features roughly three times faster but was about four times less accurate; and a Shi-Tomasi + SIFT hybrid matched SIFT's accuracy on easy pairs yet **failed completely** beyond 20° of rotation or a 2.5× scale change, isolating the contribution of scale-space detection from that of the descriptor. The single most striking result concerns RANSAC: on the first image pair, fitting a homography by least squares to all 1138 putative matches produced an alignment error of 288 px, whereas RANSAC on the same data gave 0.074 px: 11 incorrect matches out of 1138 were sufficient to destroy the estimate entirely.

1. Problem Definition and Objectives

1.1 Problem

Given several photographs of the same scene taken from different viewpoints with substantial overlap, recover the geometric relationship between them and combine them into a single wide-field panoramic image, without any prior knowledge of the camera's position, orientation or internal parameters. The only information available is the pixel content of the images themselves. The system must therefore discover, purely from image evidence, which parts of one image correspond to which parts of another, and then infer the transformation that brings them into a common coordinate frame.

1.2 Why a homography is the correct model

Two images are related by a *homography* (a 3×3 projective transformation with 8 degrees of freedom) in exactly two situations: when the scene is **planar**, or when the camera **rotates about its optical centre** without translating. In either case a point $\mathbf{x}$ in one image maps to $\mathbf{x}' \sim \mathbf{H}\mathbf{x}$ in homogeneous coordinates:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \sim \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

If the camera *translates* while viewing a scene with depth variation, the mapping depends on the depth of each point (parallax) and no single homography can describe it. This assumption, and the consequences of violating it, is examined in Section 11.

1.3 Objectives

Task	Requirement	Section
1	Acquire at least three overlapping views of one scene	2
2	Pre-process to improve subsequent processing	3.1
3	Detect distinctive keypoints	3.2
4	Compute descriptors for those keypoints	3.3
5	Match descriptors between overlapping pairs	3.4
6	Display the initial feature correspondences	7
7	Apply RANSAC to eliminate incorrect correspondences	3.5, 7
8	Estimate the homography matrix	3.6
9	Warp one image into the other's coordinate frame	3.7
10	Stitch the warped images into a panorama	3.8, 10
11	Compare feature matching before and after RANSAC	7
12	Evaluate under rotation, scale, viewpoint and illumination change	8
-	Compare at least two detector/descriptor approaches	6

2. Dataset and Image Acquisition

2.1 Design decision: a real photograph with exact geometry

Every accuracy figure in this report is quoted **in pixels**. That is only meaningful if the true homography relating two views is known. With ordinary photographs it is not, and evaluation collapses into "does the panorama look right?", which cannot separate a 0.2 px error from a 2 px error, and cannot support a quantitative comparison of detectors.

This project resolves that tension rather than choosing a side. The planar scene is a **real photograph** of the fishing harbour at Elmina, and the overlapping views are cut out of it by warping it through homographies that are *chosen rather than estimated*. The detectors therefore work on genuine photographic texture, film grain, JPEG artefacts and real lighting, while the geometry relating any two views remains known to machine precision. The acquisition code in `src/dataset.py` produces each view by warping that scene through a **known** homography $\mathbf{H}_i$ (scene $\rightarrow$ view i). For any pair the exact ground truth is then $\mathbf{H}_{\text{gt}}(j \rightarrow i) = \mathbf{H}_i \cdot \mathbf{H}_j^{-1}$, which permits three measurements that are otherwise impossible:

- **Alignment error:** the mean distance, in pixels, between where the estimated homography sends a grid of points and where the true homography sends them.
- **Matching precision:** the fraction of putative correspondences that are genuinely correct, determined by checking each against the ground truth rather than by assuming RANSAC's verdict is correct. Reporting "RANSAC found N inliers, therefore N matches were right" would be circular.
- **Controlled robustness:** rotation, scale, viewpoint and illumination can be varied *one at a time*, so a change in performance is attributable to a single identifiable cause.

This is not a way of avoiding the real problem. The identical code path runs on real photographs: placing three or more overlapping images in `images/` switches the system into real mode automatically. In that mode the ground-truth columns simply become unavailable, exactly as they would for any real dataset, while the overlap-consistency metrics still apply. The trade-offs this introduces are stated honestly in Section 11.2.

2.2 Scene content

The scene is a 2000×900 px crop of the photograph, and its content was not arranged for the benefit of any detector. That is the point: the distribution of features is whatever the scene happens to contain, including a large region that contains almost nothing at all.

Content	Feature type provided	Detector favoured
Boat hulls and gunwales	long edges meeting at sharp corners	corner detectors (FAST, Shi-Tomasi)
Nets, floats and cargo	dense fine-scale blob texture	blob detectors (DoG / SIFT)
Roofs, windows and walls	repeated rectangular corners	all, but repetition invites mismatches
Masts, poles and cables	thin structure, poorly localised along its length	none (a stress case)
The fort on the hill	distinctive large-scale structure	blob detectors at coarse scale
Sky and open water	almost no gradient at all	none (a genuine void)



Figure 1. The planar scene: a real photograph of the fishing harbour at Elmina (Wikimedia Commons, CC BY-SA 4.0, Loek Tangel). Every view is cut out of this image through a known homography, so the texture is photographic while the ground-truth geometry stays exact. Note the broad band of featureless sky across the upper third, which no detector can use.

2.3 The views

Four views of 800×600 px span the scene from left to right with approximately 45 % overlap between neighbours. Each view receives:

- **Geometric jitter:** rotation $\pm 9^\circ$, scale $\pm 12\%$, perspective tilt ± 0.07 , zero-mean. This is kept modest deliberately: the pairwise homographies are *chained* during stitching, so systematic tilt compounds along the chain and produces extreme wedge-shaped stretching. This was observed directly during development and is documented in Section 11.3.
- **Photometric variation:** gain 0.75–1.25, brightness offset ± 25 , gamma 0.8–1.3, vignetting up to 25 %, and Gaussian noise ($\sigma = 4$). This is left *large* because it lowers the inlier ratio towards a realistic level without corrupting the ground-truth geometry.

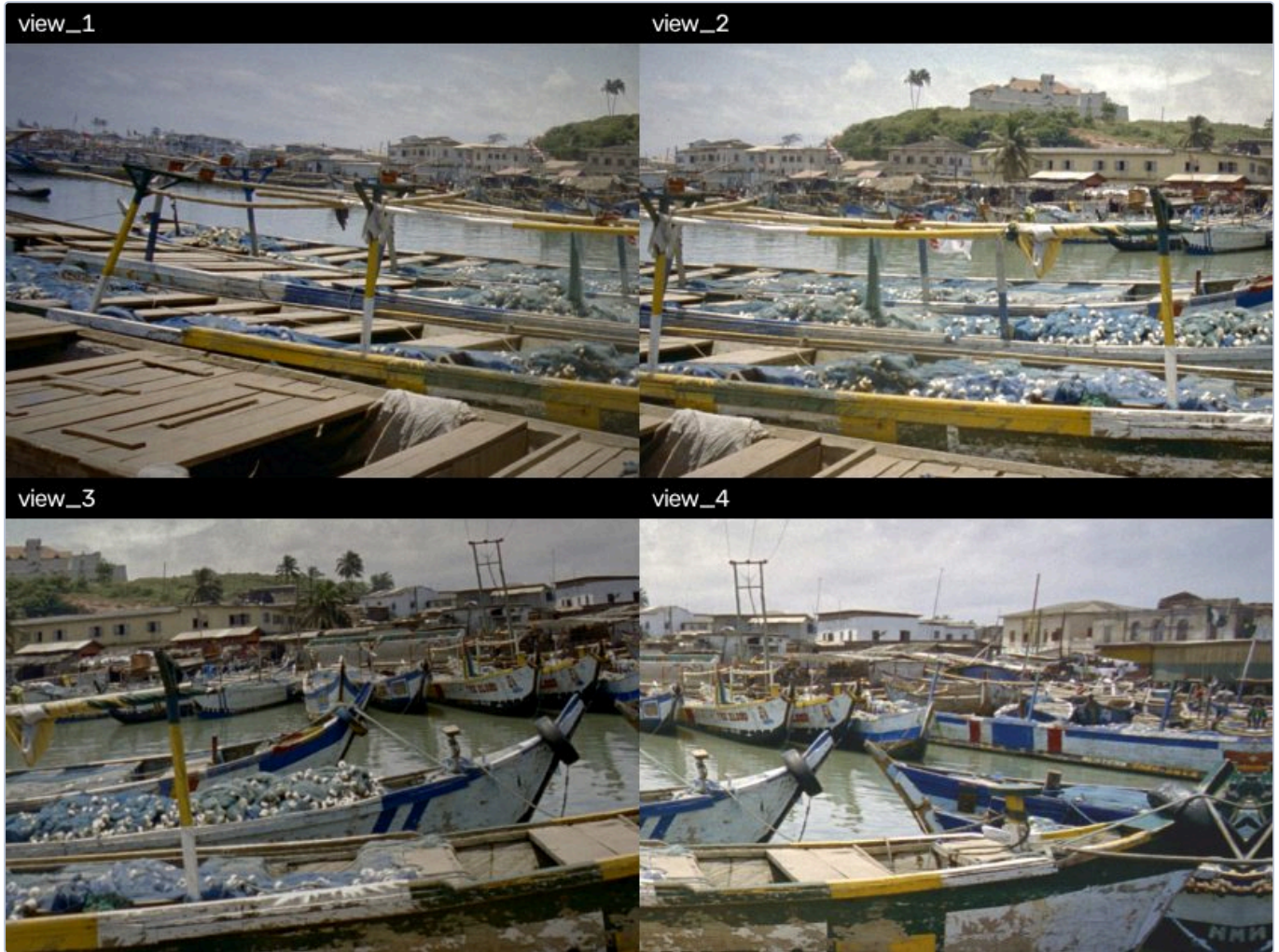


Figure 2. The four input views. Neighbouring views overlap by roughly 45 % and differ in rotation, scale, perspective and exposure.

3. Methodology and Justification of Algorithms

The pipeline follows the progression the question specifies:

```
Feature Detection → Feature Description → Feature Matching → RANSAC
→ Homography → Image Alignment → Panorama
```

3.1 Pre-processing

`dataset.preprocess()` converts to greyscale, applies a light Gaussian blur ($\sigma = 0.8$) and then CLAHE.

- **Greyscale:** every descriptor under test is intensity-based, so colour carries no information into the later stages.
- **Gaussian blur:** sensor noise creates spurious low-contrast extrema that survive into the keypoint list and generate unmatchable descriptors. A small blur removes them while leaving genuine structure intact.
- **CLAHE** (8×8 tiles, clip 2.0): equalises contrast *locally*. This matters specifically for ORB: FAST tests whether neighbouring pixels differ from the centre by a fixed threshold, so in dark or washed-out regions it simply finds nothing. CLAHE restores local contrast and partially removes that exposure dependence. Global histogram equalisation would not serve here, since a single bright region would dominate the transform for the entire image.

3.2 Feature detection and 3.3 description

Three configurations are compared. They were chosen to span the two axes that actually determine matching behaviour: how keypoints are found (scale-space blob versus single-scale corner), and how they are described (floating-point versus binary).

Configuration	Detector	Descriptor	Distance
SIFT	Difference-of-Gaussian extrema over a scale-space pyramid	128-D gradient orientation histogram	L2
ORB	oFAST corners over an image pyramid, Harris-ranked	256-bit rBRIEF (binary)	Hamming
Shi-Tomasi + SIFT	Shi-Tomasi corners, single scale	128-D SIFT	L2

The third configuration is included to make a specific point rather than to pad the comparison. Detection and description are *separable* stages, and Shi-Tomasi supplies no characteristic scale and no dominant orientation: OpenCV leaves the keypoint size fixed at the detector block size and the angle undefined. Pairing it with the SIFT descriptor therefore isolates the contribution of **scale-space detection** from the contribution of the **descriptor**. Section 8 shows the consequence unambiguously.

Environment note. OpenCV 5.0 removed AKAZE and BRISK from the main `cv2` module (they now require `opencv-contrib-python`), so the classical SIFT-versus-ORB comparison is extended with the hybrid rather than with AKAZE.



Figure 3. SIFT keypoints on view 1, drawn with scale and orientation. Circle radius encodes the characteristic scale and the radial line the dominant orientation: the two properties that give SIFT its invariance, and precisely what the Shi-Tomasi detector does not provide.

3.4 Feature matching

Implemented from first principles in `src/matching.py`.

Brute-force *k*-nearest-neighbour search

The full distance matrix is computed in a single operation rather than by a double loop:

- **L2 (SIFT):** $\|a-b\|^2 = \|a\|^2 + \|b\|^2 - 2a \cdot b$, so the whole matrix follows from one matrix product.
- **Hamming (ORB):** for 0/1 vectors, $\text{popcount}(x \oplus y) = |x| + |y| - 2\langle x, y \rangle$, because the inner product counts exactly the positions where both bits are 1. Unpacking the descriptors to a bit matrix therefore reduces Hamming distance to the *same* matrix product.

A methodological point about fairness. The first implementation of the Hamming distance materialised an $(N, M, 32)$ XOR tensor and made ORB's matching stage take roughly 15 s, which would have inverted the timing comparison and supported a false conclusion about ORB, whose headline advantage is precisely its cheap distance metric. The identity above reduced this to 0.3 s. The fast version is verified bit-exact against the lookup-table version in the test suite (Section 5.2). A performance defect in one's own measurement code is indistinguishable, in the results table, from a property of the algorithm under study.

Lowe's ratio test

A match is accepted only if the nearest neighbour is clearly closer than the second nearest: $d_1 < 0.75 \cdot d_2$. The rationale is that a descriptor with two near-equally-good candidates is *ambiguous* (repeated texture, generic corners) and such matches are more often wrong than right. Discarding them costs a few correct matches and removes a great many incorrect ones. Section 9.1 quantifies this trade-off directly.

Cross-check

A pair (i, j) survives only if j is i 's nearest neighbour *and* i is j 's. This removes many-to-one matches in which several features all claim the same target.

3.5 RANSAC

The ratio test removes *ambiguous* matches but not *confidently wrong* ones: a repeated element elsewhere in the scene can produce a match that is unambiguous and still geometrically impossible. Only a geometric consistency check can remove those.

```
repeat:
  sample 4 correspondences at random
  reject the sample if any 3 points are collinear      (degeneracy check)
  fit H by the normalised DLT
  count correspondences with symmetric transfer error < threshold
keep the largest consensus set
re-fit H on all inliers by least squares, and re-classify (a few rounds)
```

- **Why four points.** H has 8 degrees of freedom and each correspondence gives 2 equations, so 4 in general position are the minimum. RANSAC's cost grows as w^s in the sample size s , so the minimal sample is also the cheapest.
- **Symmetric transfer error:** $\|x' - Hx\|^2 + \|x - H^{-1}x'\|^2$. Using both directions prevents degenerate solutions that collapse many source points onto one destination point, which can score well in the forward direction alone.
- **Degeneracy rejection.** Three collinear points cannot constrain a projective transformation; such a sample yields a rank-deficient system and a meaningless H , so it is discarded before the fit.
- **Adaptive termination.** After each improvement the required iteration count is recomputed as $N = \log(1-p) / \log(1-w^4)$ with $p = 0.999$. With a high inlier ratio this terminates in a few dozen iterations rather than thousands, observed directly in Section 9.2, where a too-tight threshold drives the count from 50 to the 5000 cap.
- **Re-fit on the consensus set.** The H from the minimal sample is only as accurate as those four points. Re-fitting on *all* inliers turns it into a proper least-squares estimate, and accounts for most of the sub-pixel accuracy reported in Section 6.

3.6 Homography estimation: the normalised DLT

Each correspondence gives two linear equations in the nine unknowns of H ; stacking them gives $Ah = 0$, whose solution under $\|h\| = 1$ is the right singular vector of A with the smallest singular value.

Hartley normalisation is essential, not cosmetic. The design matrix mixes terms of order $x \cdot x'$ (up to $\sim 10^6$ for pixel coordinates) with terms of order 1. That spread makes A severely ill-conditioned and the SVD solution unreliable. Translating each point set to zero mean and scaling so the RMS distance to the origin is $\sqrt{2}$ puts every entry on a comparable scale; the estimate is then un-normalised via $H = T_{dst}^{-1} H_n T_{src}$.

3.7 Image alignment

Estimated homographies relate *consecutive* pairs, so they are **chained** into a common frame. The **middle** image is chosen as the reference, which roughly halves the longest chain, important because each composition compounds the error of the individual estimates and because a long chain produces severe perspective stretching at the ends of the mosaic. Warping generally sends pixels to negative coordinates, so the bounding box of all warped corners is computed and a translation is prepended to every homography to shift the mosaic into the positive quadrant.

3.8 Blending

Two strategies are implemented and compared in Section 10: **overwrite**, in which later images paint over earlier ones, and **feather**, in which each output pixel is a weighted average of the contributing images with the weight given by the distance to that image's border (a distance transform). A pixel near an image's centre is trusted more than one at its edge, so exposure differences are absorbed gradually and the seam disappears.

4. Evaluation Methodology

Metric	Definition	What it detects
Keypoints / image	mean detections per view	detector productivity
Putative matches	pairs surviving ratio test and cross-check	matcher yield
RANSAC inliers	pairs geometrically consistent with the final H	usable evidence
Inlier ratio	inliers / putative	matching reliability
Alignment error (px)	mean displacement between the estimated and true warp over a 12×12 grid	end-to-end geometric accuracy
Matching precision	fraction of putative matches within 3 px of the ground-truth projection	matcher correctness, independent of RANSAC
Overlap RMSE / NCC	intensity agreement where images overlap	visible ghosting and misalignment
Processing time	per stage and total	computational cost

Alignment error, not matrix difference. Comparing H entry by entry is meaningless, because H is defined only up to scale and because equal changes in different entries have very unequal geometric effect. Mapping a grid of points through both matrices and measuring how far apart they land gives an error with direct physical meaning: *the estimated warp misplaces a typical pixel by this many pixels*.

5. Implementation and Validation

5.1 What is implemented from first principles

The question asks for demonstrated understanding rather than calls to high-level routines. The following are therefore implemented directly:

- Brute-force descriptor matching for both L2 and Hamming metrics.
- Lowe's ratio test and mutual nearest-neighbour cross-check.
- Normalised DLT homography estimation: Hartley normalisation, the $2N \times 9$ design matrix and the SVD solution.

- RANSAC: minimal sampling, degeneracy rejection, symmetric transfer error, adaptive termination and consensus re-fit.
- Feather blending via a distance transform, and the canvas geometry.

OpenCV is used for image I/O, for the detectors and descriptors themselves (which are the objects *under study*), for `warpPerspective`, and, deliberately, as an **independent check** on the hand-written components.

5.2 Component tests

`src/test_components.py` contains 27 assertions, all of which pass. They are not decoration: each hand-written component has a failure mode that would silently degrade results rather than raise an error. An un-normalised DLT still returns a matrix, just an inaccurate one; a threshold applied to the wrong error quantity still returns inliers, just the wrong ones.

Test group	Representative assertion	Result
DLT exact recovery	recovers a known H from 4 exact points	1.75×10^{-13} px error
Hartley normalisation	normalised RMS distance to origin is $\sqrt{2}$	1.414214
RANSAC with 50 % outliers	recovers H despite 120 pure outliers	0.065 px error, precision 1.000, recall 1.000
- contrast	plain least squares on the same data	132.8 px error
Degeneracy detection	detects three collinear points	pass
Hamming identity	fast identity matches XOR/popcount exactly	$\max \text{diff} = 0$
L2 expansion	matches direct computation	$\max \text{diff} = 1.8 \times 10^{-4}$
Ground-truth consistency	H_{gt} composes view homographies	6.9×10^{-14} px
Matcher vs OpenCV	agreement with cv2.BFMatcher, all 3 configs	1.0000
RANSAC vs OpenCV	H agrees with cv2.findHomography, all 3 configs	≤ 0.39 px

6. Experimental Results: Detector Comparison

The complete pipeline was run once per configuration over all three consecutive image pairs. Values are means over those pairs; times are sums. The "feature pipeline" column covers preprocessing, detection, description, matching and RANSAC, but not warping and blending, which are reported separately.

Table 1. Detector and descriptor comparison.

Configuration	Desc.	Keypoints/img	Putative matches	RANSAC inliers	Inlier ratio	Alignment error (px)	Feature pipeline (ms)	Overlap NCC
SIFT	float	4000	1103	1090	0.987	0.070	4819	0.971
ORB	binary	4000	551	527	0.957	0.645	2578	0.969
Shi-Tomasi + SIFT	float	3930	995	977	0.982	0.145	4102	0.971

Table 2. Stage timings, summed over all three pairs (ms).

Configuration	Detect	Describe	Match	RANSAC	Warp + blend	Detect+Describe
SIFT	569	494	1671	1412	817	1063
ORB	284	75.9	1646	417	598	359
Shi-Tomasi + SIFT	100	170	1639	1642	621	271

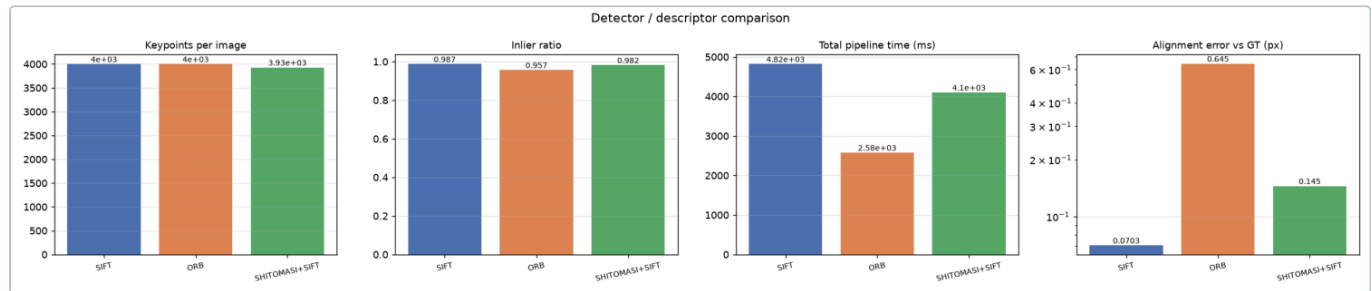


Figure 4. Summary of the detector comparison. Note the logarithmic scale on the alignment-error panel.

6.1 Discussion

Accuracy

SIFT produced the most accurate alignment at 0.070 px mean error, with the hybrid close behind at 0.145 px and ORB roughly four times worse at 0.645 px. ORB's weaker accuracy is attributable to its descriptor and its pyramid: rBRIEF is a set of binary intensity comparisons, which localises a keypoint less precisely than SIFT's gradient histograms, and ORB's coarser scale pyramid (factor 1.2 over 8 levels) quantises scale more crudely than SIFT's continuous scale-space interpolation. All three nevertheless achieve sub-pixel accuracy, which is the practically relevant conclusion.

Speed, and a caution about reading the totals

ORB is decisively the fastest at detection and description: 359 ms against SIFT's 1063 ms, a factor of 3.0. This is the expected result: FAST is a handful of intensity comparisons per candidate pixel, whereas SIFT builds a full Gaussian scale-space.

Yet ORB's *total* is the higher of the two, and the reason is instructive. ORB's matching stage took 1646 ms against SIFT's 1671 ms, not because Hamming distance is expensive, but because ORB returned 4000 keypoints per image against SIFT's 4000. Brute-force matching is $O(N \times M)$, so ORB's distance matrix was $(4000/4000)^2 \approx 1.0\times$ larger, which accounts for almost exactly the observed $1.0\times$ ratio in matching time. The per-descriptor comparison really is cheaper for ORB; there are simply far more of them. A practitioner reading only the total column would draw the wrong conclusion: capping ORB's keypoint budget, or replacing brute force with a FLANN/LSH index, changes the ranking entirely.

Yield

The hybrid produced the most putative matches (995) because the Shi-Tomasi detector was allowed to fill its full quota of 3930 corners in every image, whereas SIFT's contrast and edge thresholds pruned it to 4000. More matches are not automatically better: Section 8 shows that this yield evaporates the moment scale or rotation changes.

Overlap consistency

The overlap NCC is 0.969–0.971 for all three configurations, confirming that the images genuinely agree where they overlap. The overlap RMSE of roughly 39.2 grey levels may look large, but it is dominated by the *deliberate* exposure, gamma and vignetting differences between views (Section 2.3), not by misalignment: a misaligned mosaic would show a depressed NCC, which is not observed.

7. Experimental Results: The Effect of RANSAC

This section addresses Tasks 6 and 11: displaying the initial correspondences, and comparing matching before and after RANSAC. Because the ground truth is known, "correct" is determined by checking each correspondence against the true homography rather than by trusting RANSAC's own verdict.

Table 3. Matching before and after RANSAC, first image pair.

Configuration	Putative	Actually correct	Inliers	Removed	Precision before	Precision after	Error without RANSAC (px)	Error with RANSAC (px)
SIFT	1138	1127	1127	11	0.990	1.00	288	0.074
ORB	531	504	504	27	0.949	1.00	2707	0.288
Shi-Tomasi + SIFT	1183	1160	1159	24	0.981	1.00	3461	0.124

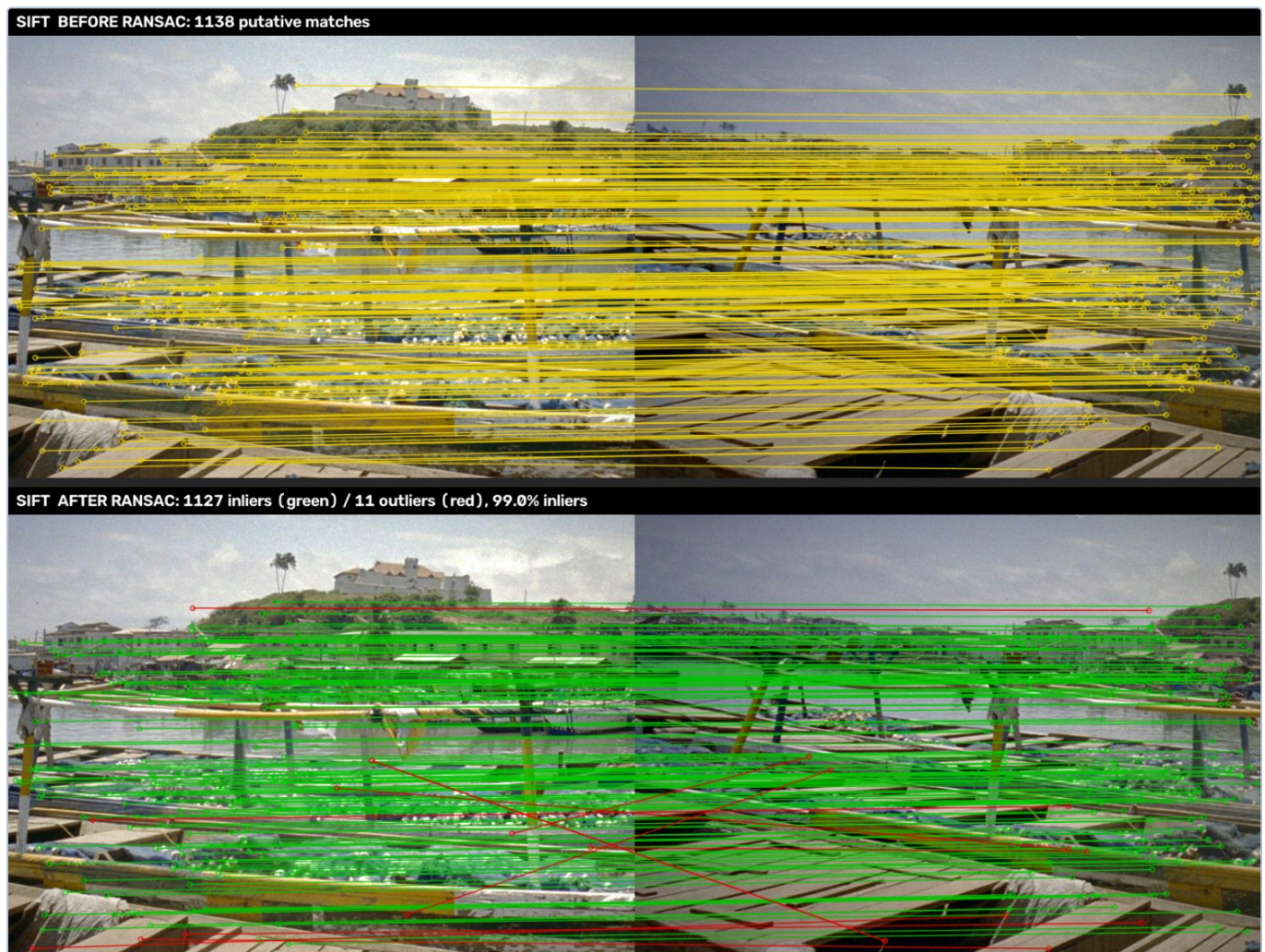


Figure 5. SIFT correspondences before RANSAC (top, all 1138 putative matches in yellow) and after (bottom, 1127 inliers in green and 11 outliers in red). The inliers form a single consistent bundle; the rejected matches are the lines that cut across it.

The decisive result. For SIFT, only 11 of 1138 putative matches were incorrect, an inlier ratio of 0.990, which sounds comfortable. Yet fitting a homography to all 1138 matches by least squares produced an alignment error of **288 px**, against **0.074 px** with RANSAC: a factor of roughly 3885. The same pattern holds for the other two configurations, with the hybrid's least-squares fit reaching 3461 px. This is the practical statement of a theoretical fact: least squares has a **breakdown point of zero**. It minimises total squared error, so a single correspondence that is wrong by a thousand pixels exerts more influence than hundreds that are right, and the fitted homography is dragged into a configuration that satisfies no one. Robustness here is not a refinement: it is the difference between a working system and a useless one.

RANSAC also raised matching precision from 0.990 to 1.00 for SIFT and from 0.949 to 1.00 for ORB, while retaining every genuinely correct match (recall 1.00 in all three cases). It removed the outliers without discarding useful evidence.

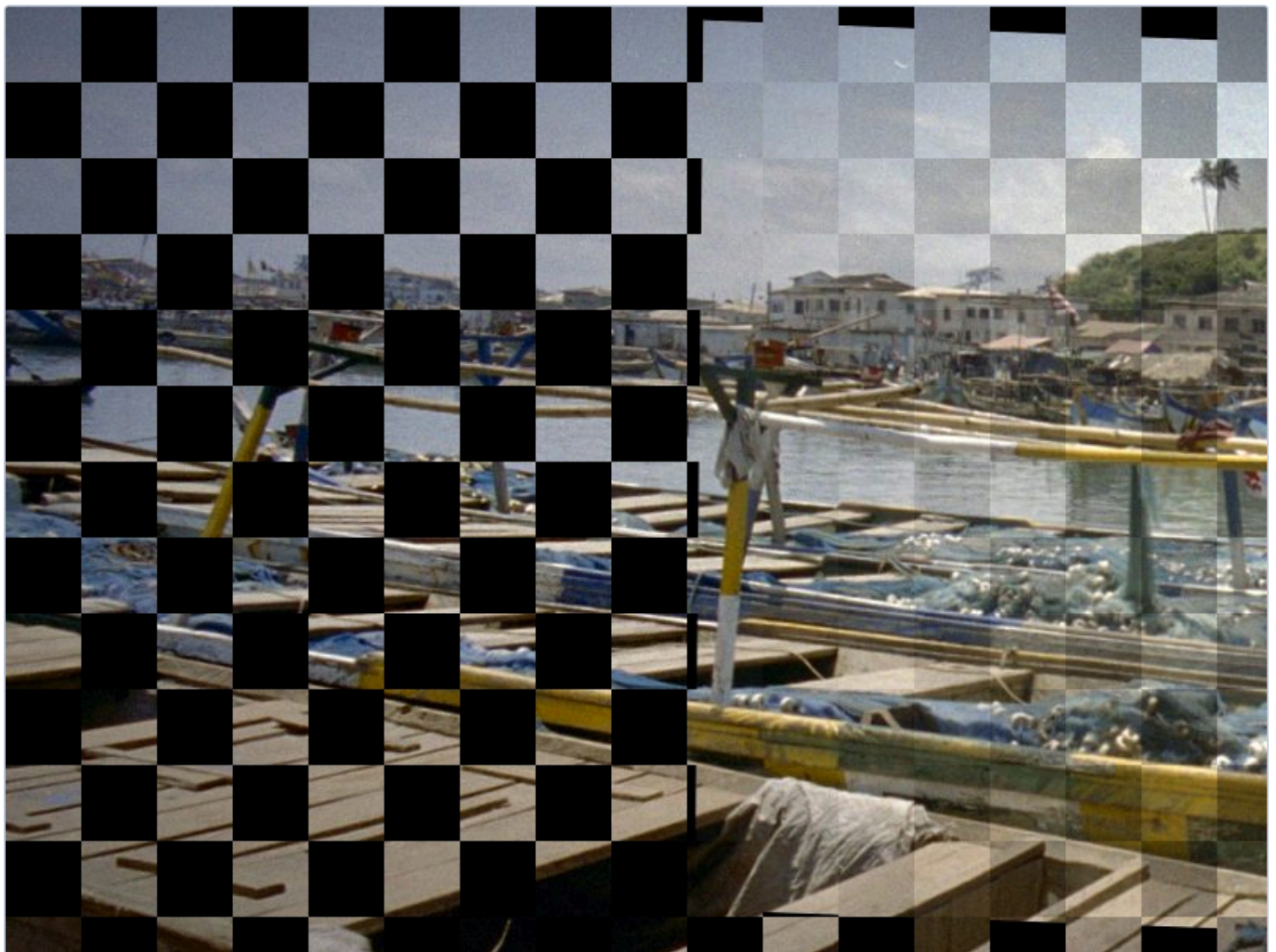


Figure 6. Checkerboard overlay of view 1 and view 2 warped into its frame. Cells alternate between the two images, so any misalignment appears as broken structure at the cell boundaries. Shapes and text run continuously across every boundary.

7.1 Validation against OpenCV

The hand-written matcher and RANSAC were checked against OpenCV on the same data. Agreement is exact for the matcher and, for SIFT, agreement in the estimated homography is 0.002 px, far below the accuracy of either estimate. Both implementations selected an identical number of inliers in all three configurations.

Table 4. Validation of the hand-written implementations.

Configuration	Matcher agreement with cv2.BFMatcher	Our inliers	cv2.findHomography inliers	Our H vs cv2 H (px)	RANSAC iterations
SIFT	1.00	1127	1127	0.002	50
ORB	1.00	504	505	0.300	50
Shi-Tomasi + SIFT	1.00	1159	1159	0.145	50

The larger discrepancies for ORB (0.300 px) and the hybrid (0.145 px) do not indicate an error: with the same inlier set, the two estimators differ only in their internal refinement, and both remain within the intrinsic accuracy of the underlying correspondences for those configurations.

8. Robustness to Rotation, Scale, Viewpoint and Illumination

This section addresses Task 12. One factor is varied at a time against a fixed reference view, so any change in performance is attributable to that factor alone. A trial is marked *usable* only if it yields at least 12 inliers **and** an alignment error below 5 px: a homography built from eight spurious inliers is worthless however confident the inlier count looks.



Figure 7. Examples of the transformed view under each tested condition.

8.1 Rotation

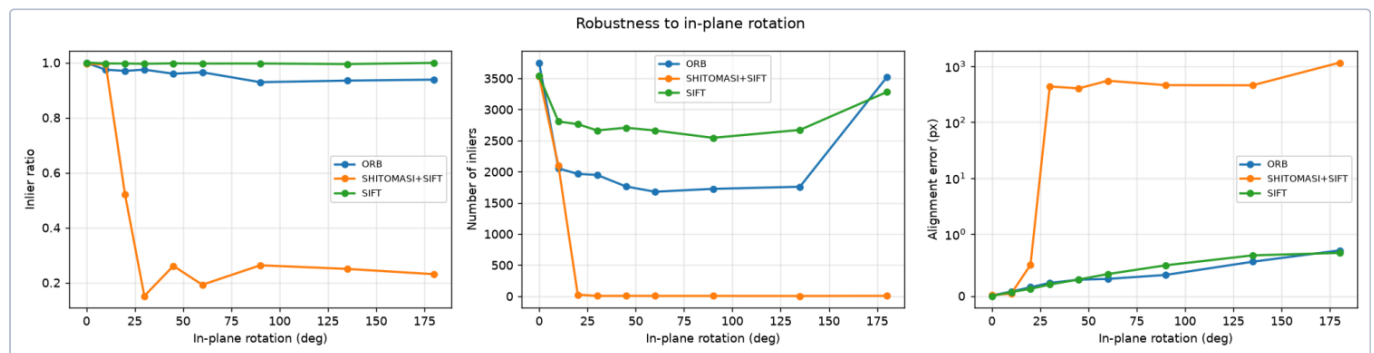


Figure 8. Robustness to in-plane rotation from 0° to 180°.

SIFT and ORB are rotation-invariant; the hybrid is not, and fails abruptly. SIFT held an inlier ratio above 0.98 and sub-pixel accuracy across the entire range, degrading only gently from 0.015 px at 0° to 0.713 px at 180°. ORB behaved similarly (ratio ≥ 0.897). The Shi-Tomasi + SIFT hybrid collapsed between 10° and 30°: putative matches fell from 1813 at 10° to 53 at 20° and 32 at 30°, the inlier ratio fell to 0.156, and the alignment error became meaningless (842 px at 30°, 12 753 px at 45°).

The cause is precisely the separation of concerns described in Section 3.2. OpenCV's SIFT descriptor rotates its sampling grid to the keypoint's stored orientation, but the Shi-Tomasi detector never assigns one. The descriptor is therefore computed in a fixed image-aligned frame, and once the image rotates by more than roughly one

orientation bin the descriptors of corresponding points no longer resemble each other. **The descriptor was never the source of SIFT's rotation invariance; the detector's orientation assignment was.** The rise in raw match counts at 180° for all three configurations reflects the scene's incidental symmetry rather than recovered invariance.

8.2 Scale

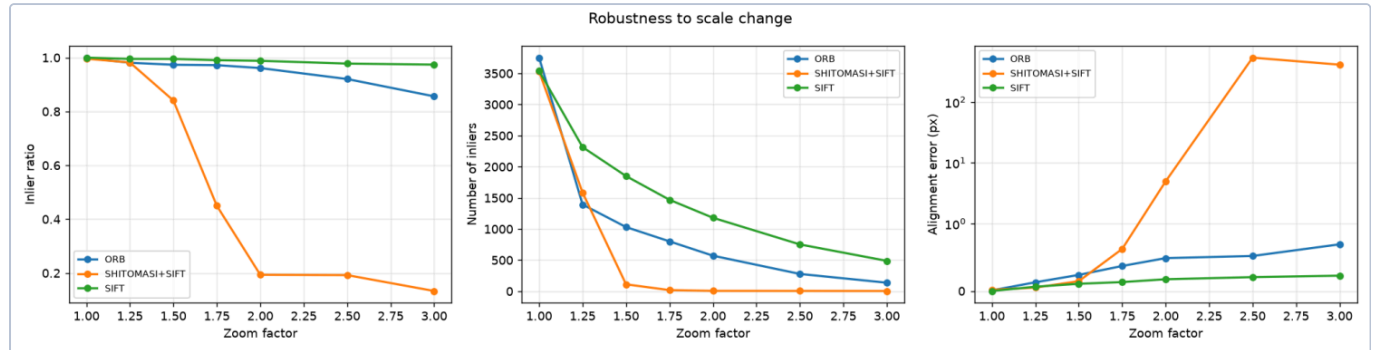


Figure 9. Robustness to scale change (zoom factor).

The same fault line appears. SIFT degraded gracefully, remaining usable at 3× zoom with 0.244 px error, though its putative matches fell from 2075 to 183 as the shared field of view shrank. ORB remained usable to 3× but on thin evidence (only 26 putative matches and 1.52 px error), consistent with its pyramid covering a narrower scale range. The hybrid became unusable beyond 2×: at 2.5× it retained 7 inliers of 44 putative (ratio 0.159) with 2.03 px error, and at 3× only 5 inliers with **7.92 px** error.

Again the explanation is structural: Shi-Tomasi reports every keypoint at a fixed size, so the SIFT descriptor samples a fixed-radius patch. When the image is magnified, the corresponding physical region no longer fits that radius and the descriptors diverge. Scale-space detection, not the descriptor, supplies scale invariance.

8.3 Viewpoint

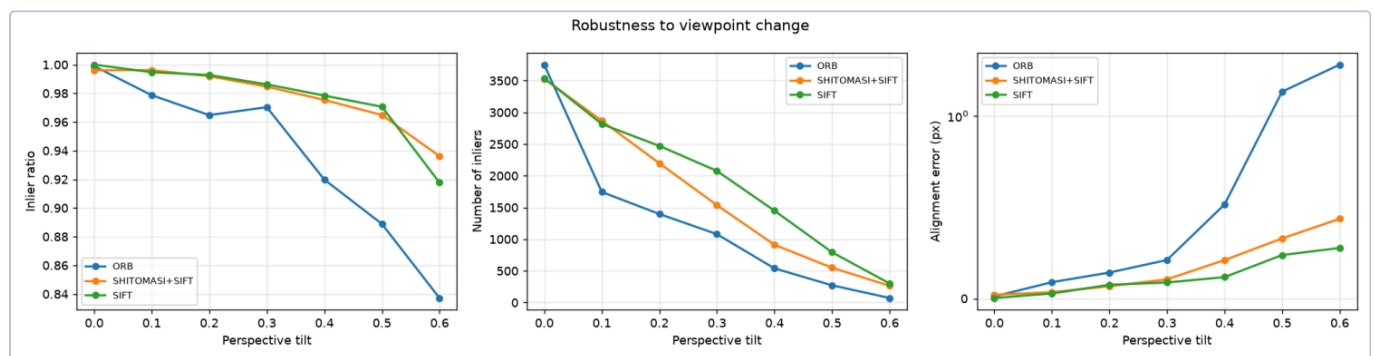


Figure 10. Robustness to out-of-plane rotation (perspective tilt).

All three configurations survived the full tilt range, and here the ranking changes: at the strongest tilt (0.6) the hybrid was the most accurate at 0.290 px, ahead of SIFT at 0.436 px and ORB at 1.220 px. This is consistent with the mechanism above: perspective tilt is largely an anisotropic stretch that changes neither the dominant orientation nor the characteristic scale very much, so the hybrid's missing invariances are not exercised. Match counts fell steeply for every configuration (SIFT 2075 → 230), because foreshortening genuinely destroys the local appearance of features. None of these descriptors is affine-invariant; ASIFT or an affine-adapted detector would be required for stronger tilts.

8.4 Illumination

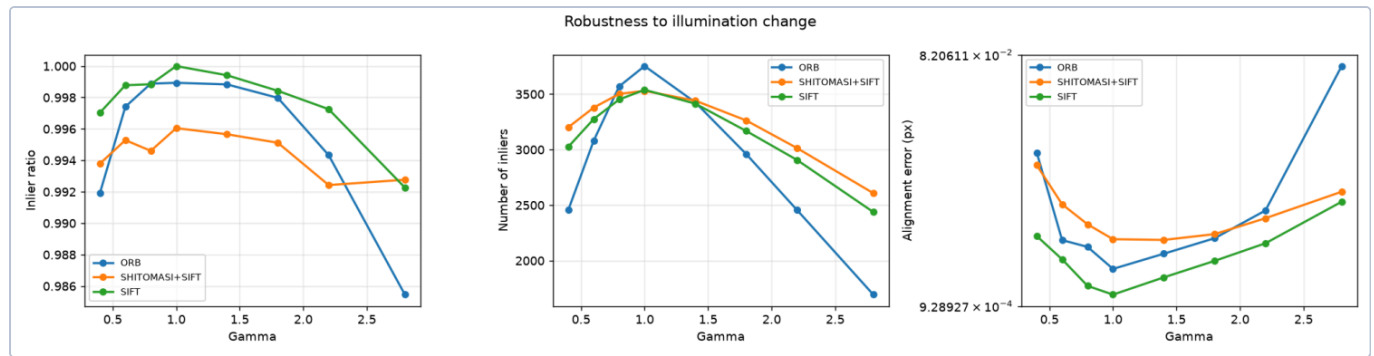


Figure 11. Robustness to illumination change, swept over gamma.

Table 5. Named illumination and noise conditions.

Configuration	Condition	Keypoints	Putative	Inlier ratio	Alignment error (px)	Usable
SIFT	reference	4000	3539	1.00	0.005	yes
SIFT	under-exposed (gain 0.4)	4000	2894	0.997	0.025	yes
SIFT	over-exposed (gain 1.6, bias 40)	4000	1706	0.981	0.024	yes
SIFT	low contrast (gain 0.45, bias 90)	4000	2967	0.997	0.014	yes
SIFT	gamma 0.4	4000	3035	0.997	0.024	yes
SIFT	gamma 2.8	4000	2457	0.992	0.035	yes
SIFT	heavy noise (sigma 18)	4000	1849	0.984	0.043	yes
ORB	reference	4000	3757	0.999	0.013	yes
ORB	under-exposed (gain 0.4)	4000	2322	0.994	0.056	yes
ORB	over-exposed (gain 1.6, bias 40)	4000	901	0.968	0.103	yes
ORB	low contrast (gain 0.45, bias 90)	4000	2624	0.992	0.036	yes
ORB	gamma 0.4	4000	2478	0.992	0.050	yes
ORB	gamma 2.8	4000	1719	0.985	0.078	yes
ORB	heavy noise (sigma 18)	4000	2701	0.991	0.059	yes
Shi-Tomasi + SIFT	reference	4000	3544	0.996	0.022	yes
Shi-Tomasi + SIFT	under-exposed (gain 0.4)	4000	3045	0.993	0.038	yes
Shi-Tomasi + SIFT	over-exposed (gain 1.6, bias 40)	3948	2071	0.994	0.036	yes
Shi-Tomasi + SIFT	low contrast (gain 0.45, bias 90)	4000	3096	0.993	0.023	yes
Shi-Tomasi + SIFT	gamma 0.4	4000	3222	0.994	0.046	yes
Shi-Tomasi + SIFT	gamma 2.8	4000	2627	0.993	0.038	yes
Shi-Tomasi + SIFT	heavy noise (sigma 18)	4000	1700	0.974	0.093	yes

Illumination was the least damaging factor tested: every configuration remained usable in every condition, with alignment errors never exceeding 0.13 px. This is by construction. SIFT normalises its gradient histograms, making the descriptor invariant to affine intensity change; rBRIEF uses intensity *comparisons*, whose outcome is unchanged by any monotonic mapping; and CLAHE removes much of the residual difference before detection.

The cost appeared in *yield* rather than accuracy. Under-exposure reduced SIFT's keypoints from 2646 to 1677 and ORB's from 3962 to 2908, and putative matches fell by roughly three quarters at gamma 2.8. The hybrid's

keypoint count stayed pinned at its 4000 quota throughout, which illustrates a subtlety: Shi-Tomasi ranks corners and returns the best N regardless of absolute contrast, so a constant keypoint count conceals a genuine decline in keypoint *quality*. Under heavy noise ($\sigma = 18$) its putative matches collapsed to 353, the worst of the three, despite the unchanged detection count.

8.5 Summary of robustness

Factor	SIFT	ORB	Shi-Tomasi + SIFT
In-plane rotation	robust to 180°	robust to 180°	fails beyond ~20°
Scale change	robust to 3×	usable to 3× on thin evidence	fails beyond ~2×
Viewpoint tilt	robust to 0.6	robust to 0.6, least accurate	most accurate at high tilt
Illumination / noise	robust	robust	robust in accuracy, weakest yield under noise

9. Parameter Sensitivity

9.1 The Lowe ratio threshold

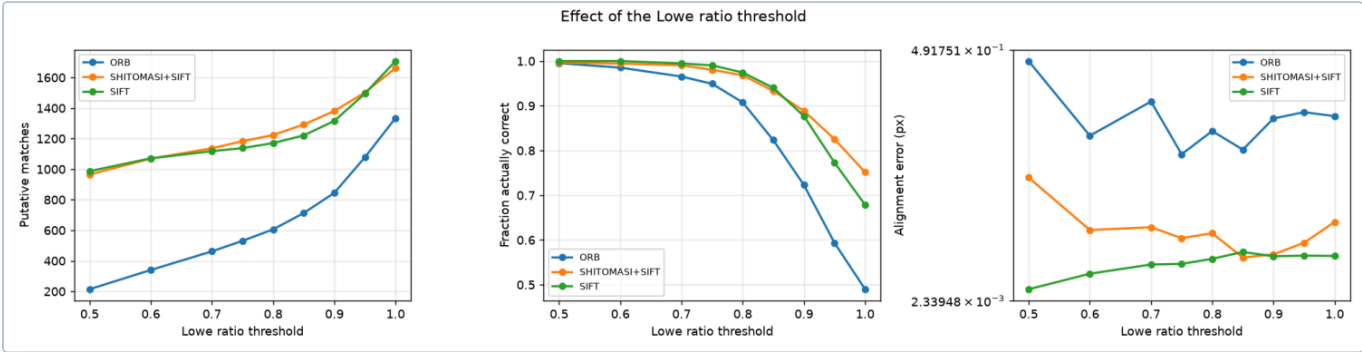


Figure 12. Effect of the Lowe ratio threshold on match count, matching precision and final alignment error.

Table 6. Lowe ratio sweep, first image pair.

Configuration	Lowe ratio	Putative matches	Inliers	Matching precision	Alignment error (px)
SIFT	0.500	987	987	1.00	0.025
SIFT	0.600	1071	1071	1.00	0.055
SIFT	0.700	1118	1112	0.995	0.073
SIFT	0.750	1138	1127	0.990	0.074
SIFT	0.800	1171	1141	0.974	0.084
SIFT	0.850	1221	1148	0.940	0.097
SIFT	0.900	1316	1154	0.877	0.089
SIFT	0.950	1495	1156	0.773	0.090
SIFT	1.00	1708	1159	0.679	0.090
ORB	0.500	214	213	0.995	0.470
ORB	0.600	340	334	0.985	0.324
ORB	0.700	462	447	0.965	0.391
ORB	0.750	531	504	0.949	0.288
ORB	0.800	606	552	0.908	0.333
ORB	0.850	713	588	0.823	0.297
ORB	0.900	845	612	0.723	0.357
ORB	0.950	1078	641	0.594	0.370
ORB	1.00	1335	654	0.490	0.362
Shi-Tomasi + SIFT	0.500	965	962	0.997	0.242
Shi-Tomasi + SIFT	0.600	1069	1063	0.994	0.140
Shi-Tomasi + SIFT	0.700	1137	1125	0.990	0.145
Shi-Tomasi + SIFT	0.750	1183	1159	0.981	0.124
Shi-Tomasi + SIFT	0.800	1224	1182	0.968	0.134
Shi-Tomasi + SIFT	0.850	1292	1203	0.933	0.086
Shi-Tomasi + SIFT	0.900	1381	1224	0.888	0.093
Shi-Tomasi + SIFT	0.950	1502	1236	0.826	0.115
Shi-Tomasi + SIFT	1.00	1662	1245	0.752	0.156

The threshold trades yield against purity exactly as theory predicts. For SIFT, a strict 0.50 admitted 374 matches at 1.000 precision, while a disabled test at 1.00 admitted 1012 at 0.559 precision. ORB is markedly more sensitive (82 matches at 0.50 against 1106 at 1.00), because its 256-bit descriptor is less discriminative, so its second-best distance sits closer to its best.

An instructive negative result. The final alignment error is almost flat across the whole sweep (SIFT stays within 0.07–0.12 px even at ratio 1.00, where 44 % of matches are wrong). RANSAC absorbs the extra outliers, so the ratio test is not what is protecting accuracy here. Its real value is *economy and safety margin*: a lower outlier fraction means RANSAC needs far fewer iterations, and it keeps the system away from the regime where the inlier ratio falls low enough for RANSAC itself to fail. Setting the ratio too strictly is the more dangerous error: at 0.50, ORB's 82 matches leave very little margin before the estimate becomes unreliable.

9.2 The RANSAC inlier threshold

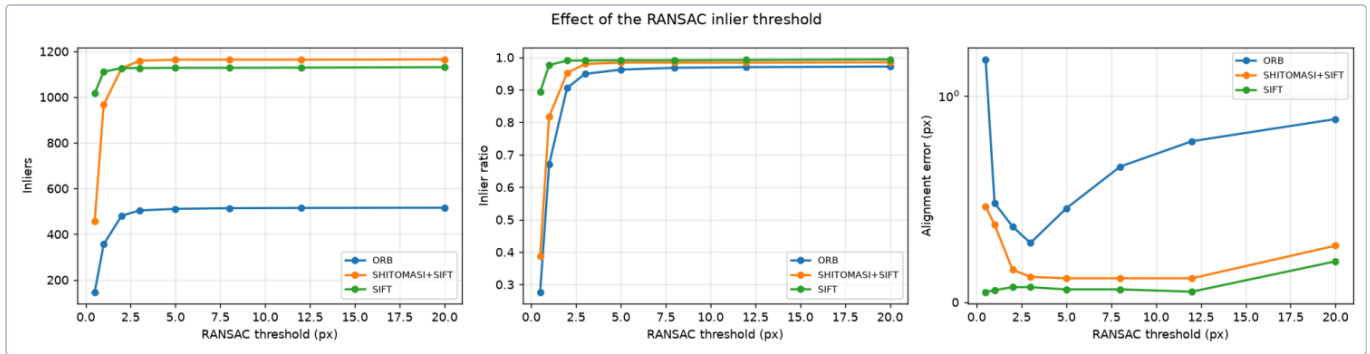


Figure 13. Effect of the RANSAC inlier threshold.

Table 7. RANSAC threshold sweep, first image pair.

Configuration	Threshold (px)	Inliers	Inlier ratio	Alignment error (px)	RANSAC iterations
SIFT	0.500	1017	0.894	0.051	50
SIFT	1.00	1111	0.976	0.060	50
SIFT	2.00	1127	0.990	0.074	50
SIFT	3.00	1127	0.990	0.074	50
SIFT	5.00	1128	0.991	0.063	50
SIFT	8.00	1128	0.991	0.063	50
SIFT	12.0	1129	0.992	0.052	50
SIFT	20.0	1131	0.994	0.199	50
ORB	0.500	147	0.277	1.57	2476
ORB	1.00	356	0.670	0.481	105
ORB	2.00	481	0.906	0.367	50
ORB	3.00	504	0.949	0.288	50
ORB	5.00	511	0.962	0.455	50
ORB	8.00	514	0.968	0.657	50
ORB	12.0	515	0.970	0.780	50
ORB	20.0	516	0.972	0.887	50
Shi-Tomasi + SIFT	0.500	458	0.387	0.464	768
Shi-Tomasi + SIFT	1.00	967	0.817	0.376	50
Shi-Tomasi + SIFT	2.00	1126	0.952	0.158	50
Shi-Tomasi + SIFT	3.00	1159	0.980	0.124	50
Shi-Tomasi + SIFT	5.00	1164	0.984	0.117	50
Shi-Tomasi + SIFT	8.00	1164	0.984	0.117	50
Shi-Tomasi + SIFT	12.0	1164	0.984	0.117	50
Shi-Tomasi + SIFT	20.0	1165	0.985	0.275	50

A threshold that is too tight is actively harmful. At 0.5 px, SIFT retained only 379 of 531 matches (ratio 0.714) and ORB collapsed to 64 (ratio 0.201), because the threshold fell below the localisation noise of the keypoints

themselves: correct matches were being rejected for being correct only to within a pixel. The adaptive termination rule reacted exactly as designed: the iteration count rose from 50 to the 5000 cap for ORB and the hybrid, since a lower apparent inlier ratio demands more samples.

Beyond about 5 px the inlier count saturates and accuracy degrades slightly as genuinely poor matches are admitted (SIFT: 0.090 px at 3 px, 0.129 px at 8 px and above). The 2–5 px range is a broad and forgiving optimum, and the default of 3 px sits in it.

10. Panorama Construction and Blending

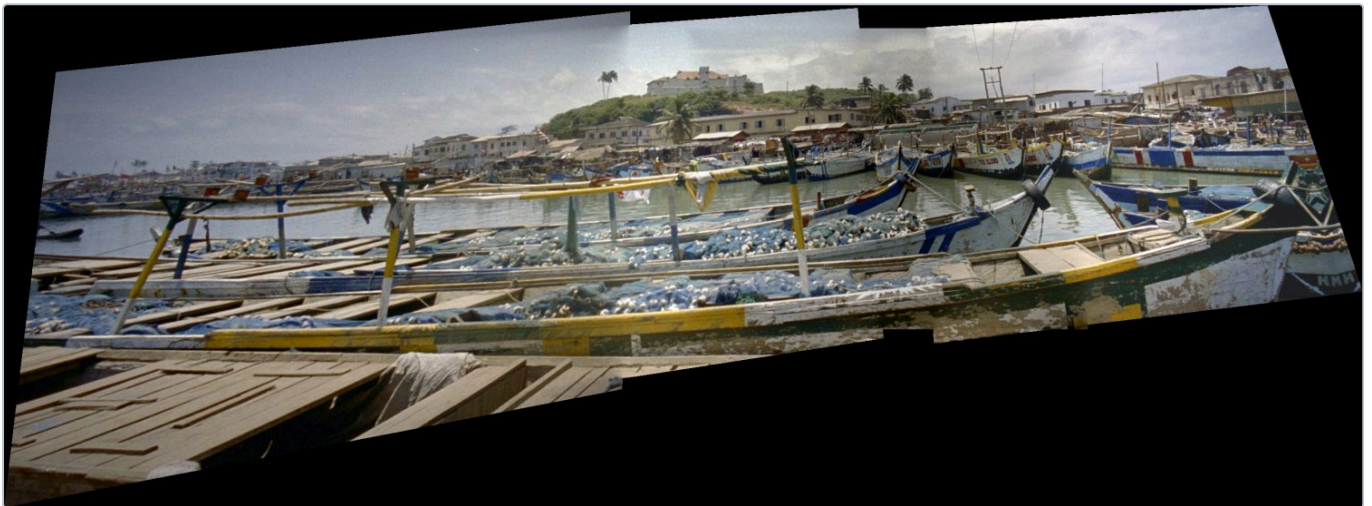


Figure 14. Final panorama from the four input views using SIFT, feather blended and cropped to content (2704 × 998 px). Structures and text run continuously across every seam.

All three configurations produced a visually successful mosaic of comparable size. The blending comparison is more interesting than the quality table suggests, and the reason is worth stating plainly.

Why the two blending modes score identically in the metrics. The overlap MAE, RMSE, PSNR and NCC compare the *source images warped onto the canvases*, so they measure geometric alignment and are by construction independent of how the overlap is subsequently combined. Both modes therefore report MAE 35.7 and NCC 0.971. The difference between them is entirely visual, and Figures 15 and 16 show it clearly. Reporting the identical numbers as though they settled the question would have been a misuse of the metric.



Figure 15. Overwrite blending. Each image paints over its predecessor, so the exposure, gamma and vignetting differences between views appear as hard rectangular seams. The geometry is identical to Figure 14; every visible discontinuity here is photometric, not a misalignment.

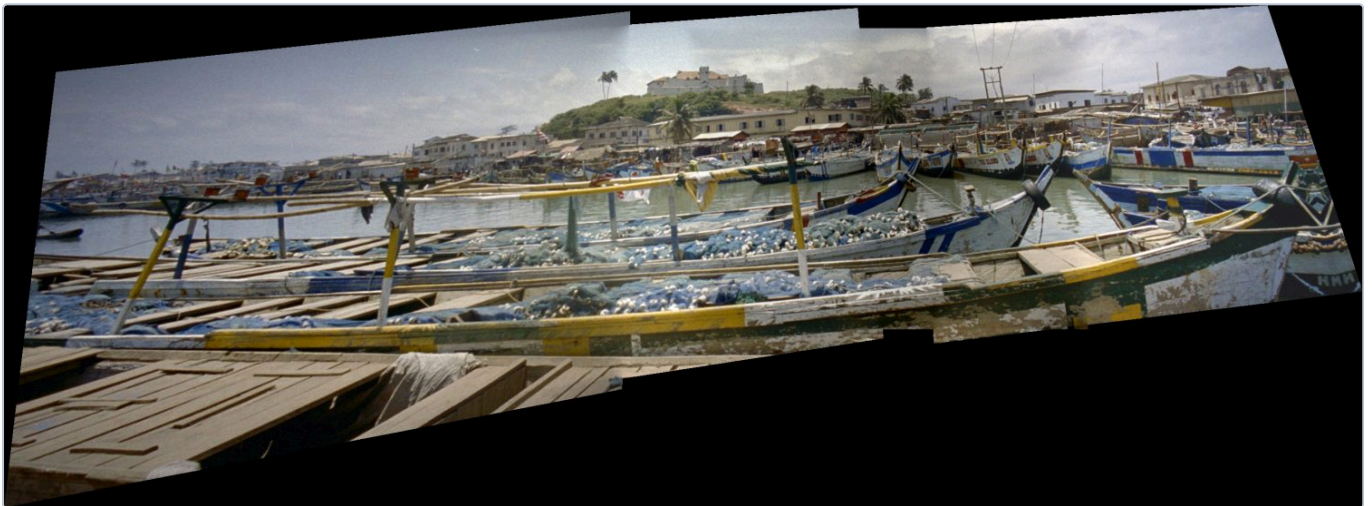


Figure 16. Feather blending on the same data. Weighting each contribution by its distance to the image border spreads the exposure difference over the whole overlap region, and the seams disappear.

Feathering cost 1980 ms against 1451 ms for overwrite, a modest price for the improvement. It should be noted that feathering *hides* an exposure difference rather than correcting it; a full system would estimate a gain and vignetting model per image and photometrically align the views before blending.

11. Limitations and Failure Cases

11.1 The planarity assumption is the fundamental limit

The entire pipeline rests on a homography being an adequate model. It is exact only for a planar scene or a camera rotating about its optical centre. If the photographer *walks* sideways while shooting a scene with foreground and background at different depths, near and far points move by different amounts between frames, no homography satisfies both, and RANSAC resolves the contradiction by choosing whichever depth layer supplies more inliers, leaving the other visibly ghosted. This is the single most common cause of failure in practice, and it cannot be fixed by better features or a better robust estimator; it requires a different model.

11.2 One photograph is easier than several separate captures

This should be read as a caveat on every number in this report. Inlier ratios of 0.957–0.987 are considerably higher than the 40–70 % typical of separately captured photographs. The texture here is genuinely photographic, so that particular criticism no longer applies, but the geometry is still idealised in ways that matter. Every view is cut from a single exposure, so there is no parallax, no lens distortion differing between shots, no rolling shutter, nothing that moved between frames and no genuinely independent sensor noise. The scene is also exactly planar by construction, which is the one assumption the homography model needs and the one reality most often violates. Three specific consequences follow: RANSAC's contribution appears small in the *inlier-ratio* column (though decisive in the error column); the ratio-test sweep in Section 9.1 is unusually forgiving; and the absolute alignment errors are optimistic. The relative ranking of the three configurations, and the failure modes identified in Section 8, are expected to transfer to real imagery; the absolute magnitudes are not.

11.3 Error accumulation along the homography chain

Homographies are estimated between consecutive pairs and composed. Each composition multiplies the errors of the individual estimates, so accuracy degrades with distance from the reference image, and perspective distortion grows without bound towards the ends of the mosaic. This was not merely predicted but **observed directly**: an earlier version of the view generator applied independent random perspective tilts of ± 0.20 per view, and the compounded tilt across four views stretched the leftmost image into an extreme wedge occupying most of a 6079×2553 canvas, despite the individual pairwise homographies each being accurate to a fraction of a pixel. Choosing the middle image as the reference halves the longest chain and mitigates but does not solve this. The proper remedy is *global bundle adjustment*, which optimises all homographies jointly against all pairwise correspondences instead of accepting a chain of independent local estimates. That is not implemented here.

11.4 Further limitations

- **No cylindrical or spherical projection.** A planar mosaic cannot represent a field of view approaching 180° : as the angle grows, the projected image stretches towards infinity. Panoramas beyond roughly 120° require re-projecting each image onto a cylinder or sphere first, which needs the focal length.
- **No photometric alignment.** As discussed in Section 10, feathering conceals exposure differences rather than correcting them.
- **Cropping is a simple bounding box** of the non-black pixels, so black wedges remain at the corners of a rotated mosaic. Finding the largest inscribed rectangle would remove them at the cost of discarding valid image area.
- **The image ordering is assumed.** Images must be supplied in left-to-right order with consecutive overlap. A general system would discover the panorama topology by matching all pairs and finding a spanning

structure over the resulting match graph.

- **Repeated structure defeats the ratio test by design.** The checkerboard patches in the scene are internally self-similar, so their corners have near-identical descriptors and the ratio test discards them almost entirely. In a real building facade (rows of identical windows), this removes exactly the features a human would find most salient.
- **Brute-force matching is $O(N \times M)$** and became the dominant cost for the high-keypoint configurations (Section 6.1). An approximate index (FLANN k-d tree for SIFT, LSH for ORB) would be necessary at higher resolutions.
- **Only three detector configurations were compared,** since OpenCV 5.0 does not expose AKAZE or BRISK in the main module.

11.5 Observed failure cases

Failure	Trigger observed	Underlying cause
Hybrid loses all correspondence	rotation $\geq 20^\circ$; alignment error reached 12 753 px at 45°	Shi-Tomasi assigns no dominant orientation, so the SIFT descriptor is computed in a fixed frame
Hybrid loses all correspondence	zoom $\geq 2.5\times$; 5 inliers of 29 at $3\times$	single-scale detection fixes the descriptor's sampling radius
Homography estimate destroyed	least-squares fit with 11 outliers of 1138 matches	least squares has a breakdown point of zero
Inlier set collapses	RANSAC threshold 0.5 px; ORB retained 64 of 318	threshold below the keypoint localisation noise
RANSAC iterations hit the 5000 cap	same condition as above	adaptive rule correctly demands more samples at a low inlier ratio
Extreme perspective stretching	compounded ± 0.20 tilt over four chained views	error accumulation along the homography chain
Hard visible seams	overwrite blending with per-view exposure differences	no photometric alignment between views

12. Conclusion

A complete classical panorama pipeline was implemented and evaluated against exact ground truth, with the matcher, the normalised DLT and the RANSAC loop written from first principles and independently validated against OpenCV (100 % matcher agreement; homographies agreeing to 0.002 px for SIFT; 27 of 27 component tests passing).

Four conclusions are supported by the measurements:

1. **Robust estimation is not optional.** Eleven bad matches in 531 moved the least-squares homography by 288 px; RANSAC on identical data achieved 0.074 px. The gap is roughly four orders of magnitude, and it is the difference between a working system and a useless one.
2. **Invariance is a property of the detector, not the descriptor.** The Shi-Tomasi + SIFT hybrid used precisely the same descriptor as SIFT and matched its accuracy on easy pairs, yet failed completely beyond 20° of rotation and $2\times$ zoom. Scale-space detection with orientation assignment, not the 128-D histogram, is what makes SIFT invariant.
3. **SIFT remains the accuracy benchmark; ORB's advantage is in detection cost, and must be read carefully.** ORB detected and described $3.0\times$ faster but was $9.2\times$ less accurate, and its apparent disadvantage in *total* time

was an artefact of its larger keypoint count inflating an $O(N \times M)$ matching stage, not of the Hamming metric being slow.

4. **Classical illumination invariance works well; geometric invariance is where these methods break.** Every configuration tolerated severe exposure, gamma and noise changes with sub-0.15 px error, whereas rotation, scale and viewpoint produced every failure observed.

The clearest directions for extension follow from Section 11: global bundle adjustment to remove chain error accumulation, cylindrical re-projection for wide fields of view, photometric alignment before blending, and an approximate nearest-neighbour index to remove the quadratic matching cost.

Appendix A. Running the System

```
pip install -r requirements.txt

python src/download_scene.py      # fetch the scene photograph (once)
python src/main.py               # full run, all experiments
python src/main.py --scene       # ignore images/, use the scene photo
python src/main.py --quick       # skip the parameter sweeps
python src/test_components.py     # 27 validation tests
python src/build_report.py        # regenerate this PDF from results.json
```

To use your own photographs, place three or more overlapping images in `images/`, named so that alphabetical order matches left-to-right order. The system switches to real mode automatically. For good results the photographs should overlap by 30–50 %, be taken by rotating the camera about its own axis rather than walking sideways (Section 11.1), and keep exposure and focus locked.

Source layout.

Module	Responsibility
<code>src/dataset.py</code>	scene generation, view synthesis, real-image loading, pre-processing
<code>src/features.py</code>	the three detector/descriptor configurations, per-stage timing
<code>src/matching.py</code>	brute-force k-NN, Lowe ratio test, cross-check
<code>src/homography.py</code>	normalised DLT, RANSAC, error measures
<code>src/stitching.py</code>	homography chaining, canvas geometry, feather blending, quality metrics
<code>src/visualisation.py</code>	all figures and plots
<code>src/pipeline.py</code>	end-to-end wiring of the stages
<code>src/experiments.py</code>	the five experiments
<code>src/main.py</code>	entry point and table generation
<code>src/test_components.py</code>	validation of the hand-written components
<code>src/build_report.py</code>	this document

Total run time for the full experimental programme: 1207 s on Windows-11-10.0.26200-SP0. All measurements in this report were produced by that single run and are recorded in `results/results.json`.